

Question

A certain halide **X** of sulfur contains halogen **Z** 70.32%. **X** can react with acetone to generate **A** and the common gas **B** that causes acid rain. **X** can also combine with a pale yellow gas **C** to obtain **D**, which contains 19.73% sulfur. **D** can also be formed by oxidation of **X** with element **E** in the presence of CsF (reaction 1). Irradiating **D** and reacting it with hydrogen produces **F**. **F** reacts with ammonia to form **G** containing a threefold axis. **D** can also react with **B** to obtain sulfone compound **H**.

X can be prepared by reacting a binary compound **I** containing 31.14% sulfur with a binary compound **J** containing 54.75% metal, while producing another sulfur-containing compound (reaction 2). **X** reacts with a boron-containing binary compound **K** to obtain substances such as **I** (reaction 3).

The following statements are correct:

- A.** All other options are incorrect
- B.** **X** and **G** have the same spatial configuration.
- C.** **A** is a nonpolar molecule
- D.** The oxidizing power of **C** is stronger than that of **E**.
- E.** **G** contains no lone pairs of electrons.
- F.** A certain product of reaction 1 can serve as a reactant in reaction 2.
- G.** The central atoms of **B** and **I** have the same hybridization.

Answer

Correct Answer: **D**

Detailed Explanation

Determine **X** and **Z**

The mass fraction of halogen **Z** in **X** is 70.32%. This means the mass fraction of sulfur is 29.68%.

We can calculate the molar mass ratio of sulfur and halogen:

$$\frac{n \times M(Z)}{a \times M(S)} = \frac{70.32}{29.68}$$

Simplified to

$$\frac{n}{a} \times M(Z) = 75.96 \text{ g/mol}$$

We try each halogen for **Z** one by one:

If **Z** is F ($M(F) = 19.00 \text{ g/mol}$), then $\frac{n}{a} \approx 75.96/19.00 \approx 4.00$. The simplest integer ratio is n=4, a=1. The chemical formula is SF_4 .

Verify the fluorine content of SF_4 : $w(F) = \frac{4 \times 19.00}{32.06 + 4 \times 19.00} = \frac{76}{108.06} \approx 0.7033$. This is very consistent with the 70.32% given in the problem.

CHECKPOINT

2 PTS

X is SF_4 , **Z** is F (fluorine)

X is SF_4 , **Z** is F (fluorine).

Determine **A** and **B**

X (SF_4) reacts with acetone ($(CH_3)_2CO$) to produce **A** and **B**. **B** is a common gas that causes acid rain.

The common sulfur-containing acid rain gas is **B** = SO_2 (sulfur dioxide).

CHECKPOINT

1 PTS

B is SO_2

SF_4 is a strong fluorinating agent that can replace carbonyl oxygen atoms with two fluorine atoms. Its complete reaction with acetone usually produces SO_2 . The reaction formula is:
 $SF_4 + 2(CH_3)_2CO \rightarrow 2(CH_3)_2CF_2 + SO_2$.

In this reaction, the organic product is 2,2-difluoropropane. So **A** = $(CH_3)_2CF_2$.

CHECKPOINT

2 PTS

A is $(CH_3)_2CF_2$

Determine **C**, **D**, **E**

X (SF_4) combines with the pale yellow gas **C** to obtain **D**. Starting from substance **D** for derivation, **C** is pale yellow, and it is speculated to be a gas formed by halogen elements, then **D** should be a sulfur hexafluoride, where the number of S is 1, so its molecular weight is $\frac{32.06}{0.1973} = 162.5g/mol$. Since it is obtained from the reaction of SF_4 , it is speculated that there are at least 4 F atoms in it, so the mass sum of the remaining two halogen atoms is $162.5 - 32.06 - 4 \times 19 = 54.43g/mol$, which happens to be one F atom and one Cl atom, so the pale yellow gas **C** is ClF .

CHECKPOINT

1 PTS

C is ClF

The reaction is: $SF_4 + ClF \rightarrow SF_5Cl$. So **D** = SF_5Cl .

CHECKPOINT

2 PTS

D is SF_5Cl

E is a simple substance. From the perspective of atom conservation, **E** must provide Cl atoms, so E is Cl_2 (chlorine gas)

CHECKPOINT

1 PTS

E is Cl_2

Determine **F** and **G**

D (SF_5Cl) reacts with hydrogen under light to produce **F**. Under light, the S-Cl bond is easier to break than the S-F bond, and the S-S bond is easier to form than the S-H bond, so the reaction can occur: $2SF_5Cl + H_2 \rightarrow S_2F_{10} + 2HCl$. So **F** = S_2F_{10} (disulfur decafluoride)

CHECKPOINT

2 PTS

F is S_2F_{10}

F (S_2F_{10}) reacts with ammonia to produce **G**. And it is known that one product has a threefold axis, and one known reaction is $S_2F_{10} + 4NH_3 \rightarrow NSF_3 + 3NH_4F + S$. So **G** = NSF_3 (sulfur trifluoride nitride)

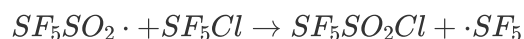
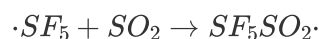
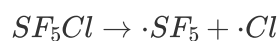
CHECKPOINT

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G is NSF_3

Determine **H**

D (SF_5Cl) reacts with **B** (SO_2) to obtain the sulfone compound **H**. One possible reaction pathway is the insertion reaction of $\cdot SF_5$ radicals:



The product **H** is SF_5SO_2Cl (pentafluorosulfanyl sulfonyl chloride), which is a sulfone compound (sulfonyl chloride is a derivative of sulfone). So

QUESTION: Determine the chemical formula of the product H. Answer: SF5SO2Cl

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H is SF_5SO_2Cl

Determine **I**, **J**, **K**

X (SF_4) can be prepared by the reaction of **I** and **J** (reaction 2). **I** is a binary compound containing 31.14% sulfur. Assuming that there is only one S atom in **I**, then:

$$w(S) = \frac{32.06}{M(I)} = 0.3114$$

That is

$$M(I) \approx \frac{32.06}{0.3114} \approx 102.95g/mol$$

I is a binary compound, set as SY_y .

If it is a sulfur oxide (SO_3 , S=40%), (SO_2 , S=50%). Not consistent.

If it is a sulfur chloride (SCl_2 , S=31.14%). $M(SCl_2) = 32.06 + 2 \times 35.45 = 102.96g/mol$. Perfectly consistent.

So **I** = SCl_2 (sulfur dichloride).

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I is SCl_2

J is a binary compound containing 54.75% metal. $S\text{Cl}_2 + \text{J} \rightarrow \text{SF}_4 + \text{another sulfur-containing compound}$. This reaction is a fluorination reaction, and **J** must be a metal fluoride. Let **J** be M_aF_n .

$$w(M) = \frac{a \times M(M)}{a \times M(M) + n \times M(F)} = 0.5475$$

◦

Solving the indeterminate equation, we get that if $n/a = 1$ (such as NaF), then $M(M) \approx 22.99$, which is element Na. So **J** = NaF (sodium fluoride). Verify the Na content in NaF: $w(\text{Na}) = \frac{22.99}{22.99+19.00} \approx 0.5475$. Consistent.

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J is NaF

X (SF_4) reacts with **K** to obtain **I** (SCl_2). **K** is a boron-containing binary compound, and from the conservation of elements, we know that **K** should also have Cl element. Therefore, there is reaction 3: $3\text{SF}_4 + 4\text{BCl}_3 \rightarrow 4\text{BF}_3 + 3\text{SCl}_2 + 3\text{Cl}_2$. This reaction can generate SCl_2 . So **K** = BCl_3 (boron trichloride).

CHECKPOINT

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K is BCl_3

Summarize all substances

X = SF_4 (sulfur tetrafluoride)

Z = F (fluorine)

A = $(CH_3)_2CF_2$ (2,2-difluoropropane)

B = SO_2 (sulfur dioxide)

C = ClF (chlorine monofluoride)

D = SF_5Cl (sulfur pentafluoride chloride)

E = Cl_2 (chlorine gas)

F = S_2F_{10} (disulfur decafluoride)

G = NSF_3 (sulfur trifluoride nitride)

H = SF_5SO_2Cl (pentafluorosulfanyl sulfonyl chloride)

I = SCl_2 (sulfur dichloride)

J = NaF (sodium fluoride)

K = BCl_3 (boron trichloride)

Analyze options

Option B:

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SF_4 is seesaw

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SNF_3 is tetrahedral

SF_4 (seesaw) and SNF_3 (tetrahedral) are different, incorrect.

Option C: $(CH_3)_2CF_2$ is a polar molecule. Incorrect.

Option D: ClF oxidizing property $\gt Cl_2$. Correct.

Option E:

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SNF_3 has 10 lone pairs of electrons

The 10 lone pairs of electrons in SNF_3 exist on nitrogen and fluorine atoms. Incorrect.

Option F: Reaction 1: $SF_4 + Cl_2 + CsF \rightarrow SF_5Cl + CsCl$

Reaction 2: $3SCl_2 + 4NaF \rightarrow SF_4 + S_2Cl_2 + 4NaCl$. Incorrect.

Option G: SO_2 (sp^2) vs SCl_2 (sp^3). Different. Incorrect.